

Transoceanic Communication by Means of Satellites

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The existence of artificial earth satellites and of very low-noise maser amplifiers makes microwave links using spherical satellites as passive reflectors seem an interesting alternative to cable or tropospheric scatter for broad-band transatlantic communication.

A satellite in a polar orbit at a height of 3000 mi would be mutually visible from Newfoundland and the Hebrides for 22.0% of the time and would be over 7.25° above the horizon at each point for 17.7% of the time. Out of 24 such satellites, some would be mutually visible over 7.25° above the horizon 99% of the time. With 100-ft diameter spheres, 140-ft diameter antennas, and a noise temperature of 20°K , 85 kw at 2000 mc or 9.5 kw at 6000 mc could provide a 5-mc baseband with a 40-dB signal-to-noise ratio.

The same system of satellites could be used to provide further communication at other frequencies or over other paths.

I. INTRODUCTION

The time will certainly come when we shall need a great increase in transoceanic electronic communications. For example, the United States and Western Europe have a wide community of interests and are bound to demand more and more communication facilities across the Atlantic. If we are to be ready to fill these growing needs, we shall have to investigate all promising possibilities.

In doing so, we shall certainly want to keep in mind a rule founded on experience. This rule is that telephone circuits become cheaper the more of them we can handle in one bundle. Then, too, there is a possibility of requirements for television. In either case, there is a premium on availability of wide bands of frequency.

The submarine cable art is presently distinctly limited in bandwidth. No doubt its capability in this respect will improve as the years go by, but we may well run into economic or technical restrictions not suffered by other techniques.

A chain of UHF scatter links over a northern route might provide channels across the Atlantic Ocean but the quality is dubious, the available bandwidth is limited, and the cost is great. Indeed, we cannot now imagine how one might improve quality of bandwidth while at the same time reducing the costs of such a system. Moreover, such links would not serve for some transoceanic routes.

An undersea millimeter wave system using a round waveguide excited in the TE₀₁ mode is a possibility for the remote future, but such a system is far beyond present technology.

A microwave system using satellite repeaters may have many advantages over the foregoing alternatives. Present rocket technology is at least close to the point of putting in orbit some structure which could act as a reflector or passive repeater. The maser amplifier, which introduces only around a hundredth the noise of earlier amplifiers, cuts down the transmitting power required to a hundredth of that arrived at in an earlier study [1]. This means that a satellite link with attractive properties could be attained within existing microwave art. The cost of a pair of microwave installations at the terminals would probably be less than the cost of a cable of far less bandwidth.

When highly reliable long-life microwave components and power supplies suited to a space environment are available, active repeaters may provide useful communication. When, in addition, accurate enough guidance is available, together with long-life means for adjusting attitude and position in orbit, a "fixed" repeater in a 24-hour orbit could be used.

Obviously, the present state of knowledge is insufficient for the design of a transatlantic satellite communication system of assured performance and cost. Much remains to be learned. For this very reason, and because they appear to be serious contenders for the future, it is important that research on satellite systems be given serious attention.

II. ALTERNATIVE SATELLITE REPEATER SCHEMES

A number of alternative types of orbital radio relays were discussed by Pierce [1]. These can be divided into classes in two essentially different ways: 1) active and passive repeaters, and 2) fast-revolving (relatively near) repeaters and repeaters in 24-hour orbits (stationary with respect to earth). The most important characteristics of these four possible types of repeater systems are summarized in Table 1.

Table 1

Orbit	Repeater	
	Passive	Active
Near (1- to 3-hour period)	Simplest embodiment. Metalized plastic sphere, 100-foot diameter. (On the ground: large-size steerable antenna.)	Carries lightweight microwave repeater and power supplies. Low-directivity antennas (On the ground: medium-size steerable antennas).
Far (24-hour period)	Plane reflector. Attitude stabilized. (On the ground: extra-large fixed antennas.)	Carries heavy microwave repeaters and power supplies. High-directivity antennas. Attitude stabilized. (On the ground: fixed medium-size and small antennas.)

It should be noted that in the case of the passive repeater the bandwidth available is almost unlimited. The passive repeater is a truly linear device, and it can be used simultaneously in many ways, at many frequencies, and with many power levels, without crosstalk. Thus, the only cost in adding new channels is that of adding terminals, and these may be of many sorts.

In contrast, active repeaters have a limited dynamic range, as well as a limited bandwidth and, hence, can be used for a limited number of separate simultaneous signals only, and the levels and natures of several signals passing simultaneously through an active repeater must be carefully controlled if they are not to jam one another.

On the other hand, smaller ground antennas and transmitter powers can be used with active repeaters, although for broad-band use large antennas and large transmitter powers have certain advantages.

Certain other considerations concerning these various possible systems will be brought out in the following sections.

III. ORBITS, MUTUAL VISIBILITY, AND DISTANCES

A. The 24-Hour Orbit

The 24-hour orbit has received considerable attention and does not need extensive discussion here. If fixed antennas, perhaps of very large size, are to be used, a satellite will have to revolve in the equatorial plane at a distance of roughly 26 000 mi from the earth's center. It is hoped that perturbations from its mean position due to the attraction of the moon and sun will be small enough so that only relatively small motions of the antenna feeds and corrections of orbital position would be required. One repeater would suffice to span the Atlantic Ocean, and one to span the Pacific. Areas of mutual visibility could be

controlled by means of the size and orientation of the reflecting surfaces, or antennas, on the satellite. With the advanced technology required for an active 24-hour repeater, it should be easy and desirable to provide communication over various paths, and so, to some degree, to overcome the limitations imposed by the fact that active repeaters can handle only a limited number of signals lying in a limited range of power levels.

Many variations on this theme are possible and will be realized in due course. The transmission provided by such a system would of course be uninterrupted.

B. Near Orbits

The near orbits, that is, orbits between 1000 and 3000 mi above the earth, can be classified into equatorial, polar, and inclined orbits. It is intuitively clear that the utility of a satellite orbit depends on the distances from the satellite to the terminal points for the portion of the orbit for which the satellite is visible from both terminal points. It is also obvious that this distance should be as small as possible, consistent with the requirement that a substantial portion of the whole orbit should be simultaneously visible from the terminal points.

Transatlantic communications appear to be of the greatest immediate importance, for the North Atlantic routes are already carrying the heaviest traffic in every available medium. Thus, we shall choose a transatlantic route as an example. Equatorial orbits, which are not suited to this route, will not be examined here. In order to determine whether inclined or polar orbits are more advantageous for transatlantic use, computations of mutual visibility have been made for two comparable cases; the result is that polar orbits are more efficient for terminations which might be chosen for a transatlantic link. Therefore, all the subsequent calculations have been made on the basis of polar orbits only.

C. Visibility Considerations

A computation has been made of the durations of simultaneous visibility of a satellite at various heights from points in North America and Europe, selected, somewhat arbitrarily, in the following locations:

Terminal A — Newfoundland	{	48° north
		55° west
Terminal B — Island of Lewis in the Hebrides, Scotland	{	58° north
		7° west.

They are about as close to each other as can be found on a map, keeping in mind that there will have to be microwave links between the terminals and the respective continental communication networks.

Assuming the radius of the earth to be 3950 mi, we find that the distance between terminals A and B is 2060 mi. For the sake of simplicity we have assumed that the satellite becomes visible as soon as it crosses the horizon and that it moves in perfectly circular orbits.

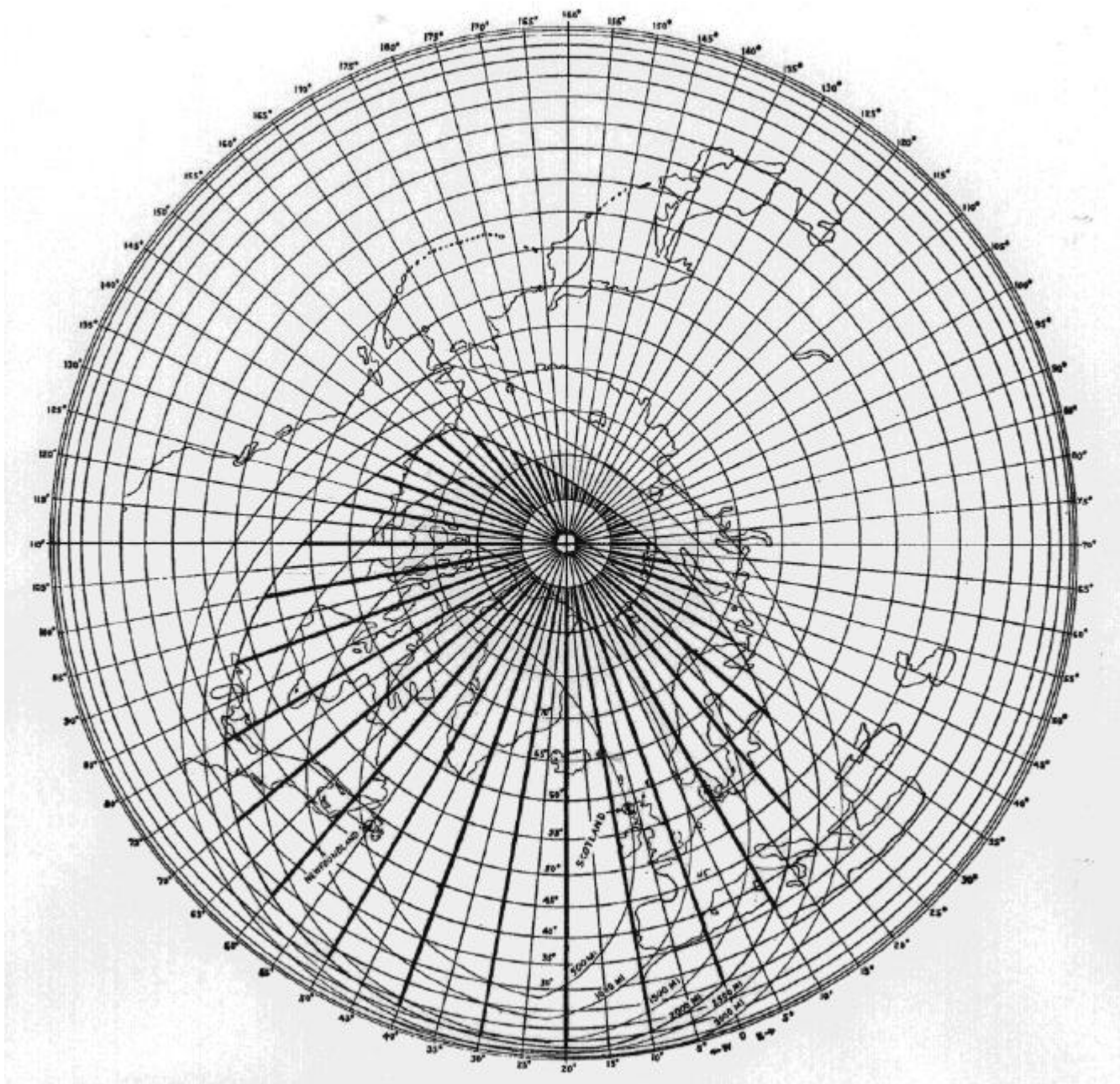


Fig. 1. Regions of mutual visibility of satellites in circular orbits at various heights. Note: For the sake of comparison with Fig. 2, polar orbits at the 2000-mi height have been darkened.

A map showing the regions of mutual visibility of satellites at heights of 500, 1000, 1500, 2000, 2500, and 3000 mi will be found in Fig. 1. These regions are projected onto the earth's surface, with the earth's center as the center of projection, and this in turn is shown in orthogonal projection, with the North Pole at the center. At the higher altitudes these regions extend close to the equator and are very much foreshortened, and another projection (not shown) has been used in computing visibility involving these regions. So far these visibility regions do not depend on the type of orbit. However, it is a simple manner to compute visibility durations for polar orbits, and this is what has been done.

The constructions of Fig. 1 have been used in calculating the length of time a satellite is visible from terminals A and B simultaneously, as a function of height h above the earth's surface and as a function of longitude of the orbit. Since, in general, the period of rotation of the earth will not be an integral multiple of the period of the satellite, the orbit will in due course occupy all possible orientations with equal probability. Therefore, an *average* percentage visibility could be calculated, an average which has been determined by summing over 72 equally spaced orbits, i.e. every 5° of longitude. The longest and shortest visibility durations have also been calculated. It will be appreciated that above a certain height (which turns out to

Table 2

	Height above surface of earth (miles)					
	500	1000	1500	2000	2500	3000
Time of one revolution (minutes)	100.4	118.0	136.0	155.0	175.2	195.2
Shortest visibility (minutes)	0	0	8.0	12.5	23.8	31.4
Longest visibility (minutes)	14.7	20.0	29.6	36.6	46.2	55.4
Average visibility (per cent)	3.5	6.9	12.9	17.7	19.6	22.0
Longest distance from A or B to satellite (miles)	2050	2980	3755	4450	5099	5718
Shortest distance from A or B to satellite (miles)	1180	1540	2100	2670	2820	3300

Assumptions: Terminals in Newfoundland and Hebrides.

Polar orbits.

Refraction effects ignored.

Visibility from horizon to horizon.

be 1350 mi for the terminations chosen) the satellite will be mutually visible from A and B for some time during every revolution, although such time may only be of short duration.

Table 2 gives the results. Note, for instance, that at a height of 2500 mi the satellite is visible every time it goes around, the shortest visibility lasting 24 min, and the longest, 46 min. With a period of revolution of about 3 h, an average visibility is about 20%.

A look at Fig. 2, showing the regions in which the satellite is visible both from A and B, now with quasi-equatorial orbits (in which the various sputniks move), shows that these orbits are not as favourable as the polar ones just discussed.

It is assumed here that the satellites are launched at an inclination so that the farthest excursions are 55° north or south. In computing the average visibility, the rotation of the earth has been neglected, as it was when computing the polar orbits. For a satellite height of 200 mi, it turns out that the average visibility is 10.2% for inclined orbits, whereas it is 17.7% for polar orbits. For equatorial orbits to yield visibility durations comparable to polar ones, terminals would have to be much nearer the equator than the ones chosen. Such locations would be farther apart from each other; moreover, they would be rather remote from the more densely populated areas of the world which possess microwave networks. Therefore, a polar orbit should be chosen.

So far it has been assumed that satellites are visible and fully utilizable right down to the horizon. There may be systems for which this is so; however, in many cases it must be assumed that satellites have to rise to a certain minimum elevation before they can be properly acquired. In particular, this will be so with systems which use maser amplifiers of high effective sensitivity (of low effective noise temperature). Propagation anomalies due to the earth's atmosphere can also be expected to be of importance very near the horizon, which is another reason for avoiding working near the horizon.

Hogg [2] has calculated the thermal noise due to the oxygen in the atmosphere. The "tail" of the well-known absorption in the 5-mm band extends far into the longer wavelengths, and thermal noise radiation is accordingly generated and will be received by any antenna in amounts

depending on the frequency and on the angle at which the beam traverses the atmosphere. Fig. 3 (based on Hogg's calculation) shows the effective temperature of an ideal antenna as a function of frequency and angle of elevation. Also shown are curves indicating the approximate limits of effective noise temperature due to cosmic noise sources.

The optimum frequency for satellite radio systems of high sensitivity will lie somewhere between 2000 and 6000 mc. We further note there is no point in making the receiver noise temperature lower than 10°K if the antenna elevation is less than 10° above the horizon. Lowering the antenna right down to the horizon increases its temperature nearly tenfold; this would hardly be noticeable with old-fashioned receivers, but would seriously degrade the performance of a receiver using a maser amplifier as the input stage.

Rather than calculate visibility regions, times, and distances afresh for various satellite heights and minimum elevations, we have taken the 3000-mi-high orbit and calculated the minimum elevations which give the same visibility regions and periods as satellites followed right down to the horizon in orbits at the following heights: 2500, 2000, and 1500 mi. The results are given in Table 3.

Comparing Tables 2 and 3, note that the price paid for using the sky only down to, say, 7¼° above the horizon is a drop in average visibility from 22.0% to 17.7%.

IV. NUMBER OF SATELLITES

One could get a great deal of useful communication with one satellite, but to provide nearly uninterrupted transmission requires many. While it is, in principle, possible to imagine satellites placed in perfectly circular orbits at regular intervals, so that they provide uninterrupted service, it is at present more realistic to assume that they will tend to appear with complete randomness. Thus sometimes there will be several visible at the same time, and sometimes there be none. The question arises: what fraction of the time, on the average, will there be none visible?

Let f be the average fraction of the satellite period when it is visible from both terminals. This is also the probability of mutual visibility, and $(1-f)$ is then the probability of the satellite *not* being seen. Hence, the probability of not seeing any of n satellites

$$(1 - f)^n.$$

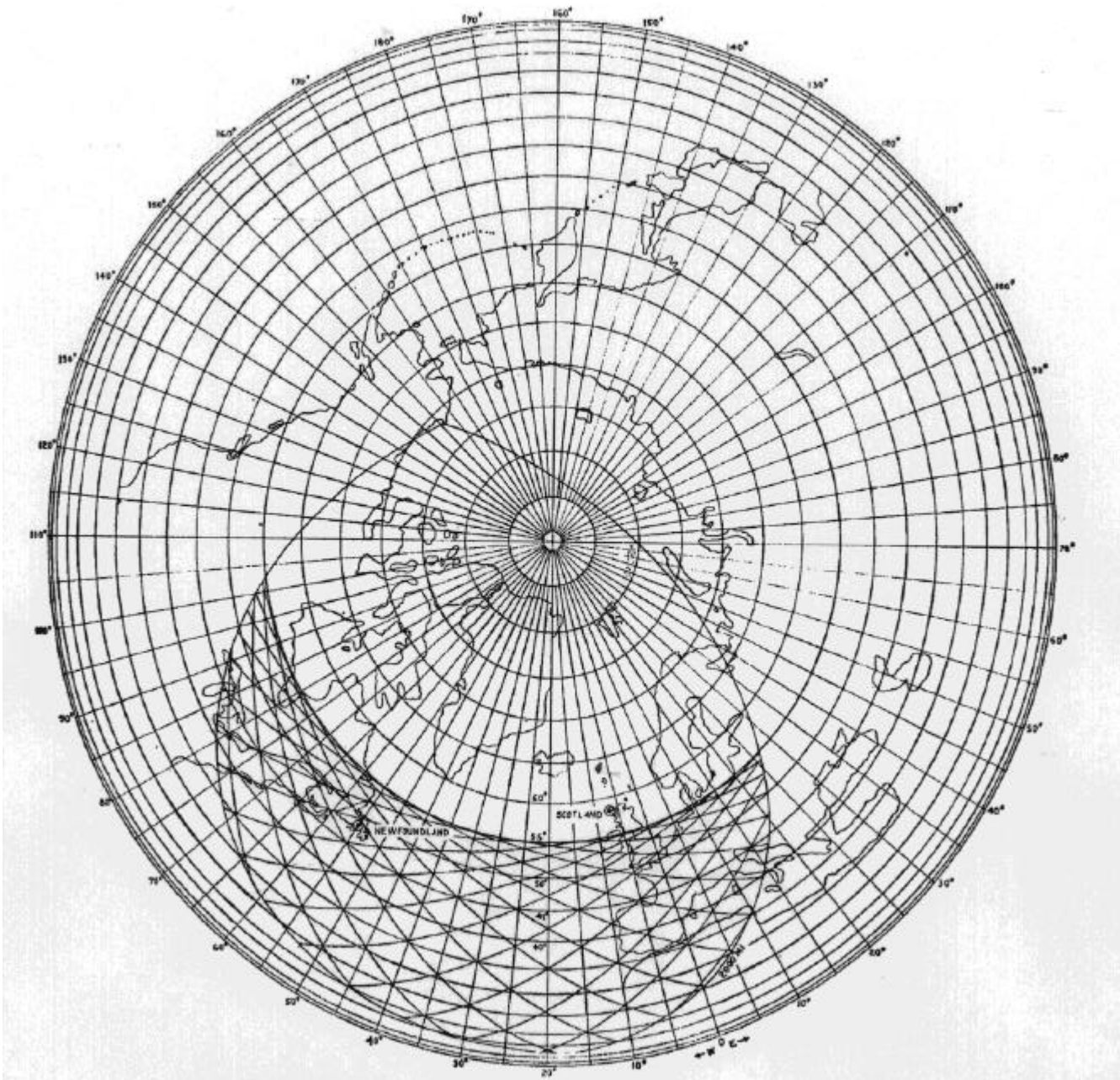


Fig. 2. Region of mutual visibility of satellites in orbits inclined 55° at a height of 2000 mi.

If this is set equal to i , the average fraction of service interrupted, the required number of satellites is obtained

$$n = \frac{\log i}{\log(1-f)}$$

Table 4 shows the minimum numbers of satellites required to give service interrupted, on the average, by the amounts indicated. This has been done for various satellite heights, assuming visibility right down to the horizon. In Table 5 there is a similar set of numbers of satellites calculated for various minimum elevations, assuming all satellites to travel at a height of 3000 mi. Note that doubling the number of satellites reduces the amount of interruption by one-tenth.

It should be noted at this point that the interruptions, though happening at irregular intervals, should be predictable well in advance, and the communication services can be organized accordingly.

V. PATH-LOSS CALCULATIONS

Many factors enter into a calculation of transmission performance of a system involving satellite repeaters between terminals on earth. Rather than carry out calculations on a variety of schemes, we shall concentrate on one in particular, namely, frequency modulation with feedback [3], which is particularly applicable to satellite communications, as has been pointed out by Ruthroff and Goodall [4]. The proposed system compares favourably with other known systems, such as PPM, PCM, or SSB. The

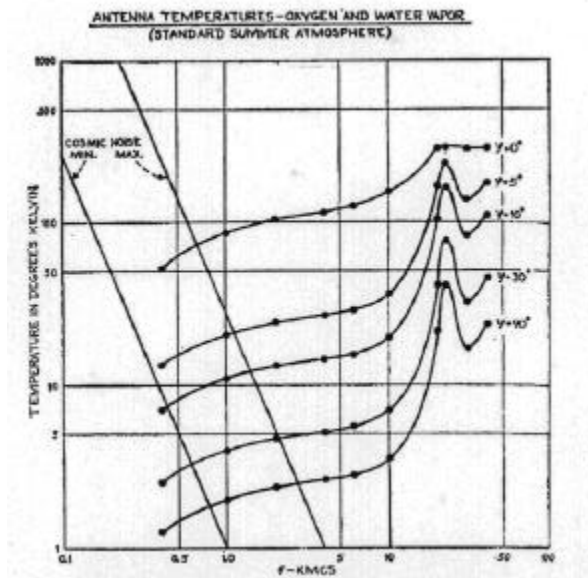


Fig. 3. Temperature due to oxygen absorption seen by an ideal antenna as a function of elevation angle ϕ and frequency.

results will be obtained in a fashion so that extrapolations to other systems can be easily performed.

Starting with the case of the passive repeater in the form of a reflecting sphere with a diameter D , the path loss can be written, as for the well-known radar case:

$$L = \frac{P_T}{P_R} = \frac{16\lambda^2 p^4}{A^2 \eta^2 D^2}$$

where

P_T = transmitter power

P_R = receiver power

λ = wavelength

p = geometric mean of the distances between the satellite and the terminals

A = antenna area (assumed to be the same for both transmitter and receiver)

η = antenna efficiency.

The noise power at the receiver input can be written

$$N = kTB$$

where

k = Boltzmann's constant (1.38×10^{-23} watt per degree per cycle per second, or -228.9 dBw for 1°K and 1 Hz)

T = effective noise temperature of receiver (including sky noise, antenna loss, tube noise, etc.) in $^\circ\text{Kelvin}$

B = the RF bandwidth in cycles per second

b = baseband modulation bandwidth.

Table 3

	Minimum elevation angle (degrees)			
	0	3.25	7.25	12.60
Average visibility (per cent)	22	19.6	17.7	12.9
Shortest visibility (minutes)	31.4	26.5	15.5	11.5
Longest visibility (minutes)	55.4	51.5	45.5	42.5
Longest distance (miles)	5718	5520	5240	4960

Assumptions: Terminals in Newfoundland and Hebrides.

Satellite height: 3000 miles.

Orbital period: 195.2 minutes.

Table 4 Number of Randomly Spaced Satellites to Provide Service Interrupted By No More than 100%

Satellite height	Number of satellites for indicated percentage of service interruption			
miles	10 per cent	1 per cent	0.1 per cent	0.01 per cent
500	62.5	125.0	187.5	2500.0
1000	32.3	64.6	96.9	129.2
1500	16.7	33.3	50.0	66.6
2000	11.8	23.6	35.4	47.2
2500	10.5	21.0	31.5	42.0
3000	9.3	18.5	27.8	37.0

Assumptions: Terminals in Newfoundland and Hebrides.

Polar orbits.

Visibility from horizon to horizon.

Table 5 Number of Randomly Spaced Satellites Required to Give Service Interrupted No More than 100%, as a Function of Minimum Elevation for Orbits at 3000-mi Height.

Minimum elevation angle	Number of satellites for indicated percentage of service interruption		
degrees	10 per cent	5 per cent	1 per cent
0	9	12	19
3.25	11	14	21
7.25	12	15	24
12.60	17	22	33

Assumptions: Terminals in Newfoundland and Hebrides Polar orbits.

In an FM receiver, noise problems are minimized if feedback is applied to reduce the deviation in the IF stages and the limiter. The operation of such a receiver is described by Chaffee [3] and Carson [5] and has been digested by Goodall and Ruthroff [4]. More recently, a closely related system has been described by Jaffe and Rehtin [6].

An optimized receiver of this kind will use sufficient feedback to reduce the deviation in the IF circuits to near zero. Under these conditions

$$(S/N) \approx 3 \left(\frac{C}{N} \right) M^2$$

and

$$B \approx 2(1 + M)b$$

where

M = modulation index

(S/N) = signal-to-noise ratio at the receiver output

(C/N) = signal-to-noise ratio at the IF frequency.

To give reliable operation

$$\left(\frac{C}{N}\right) \geq 16$$

which is the usual 12-dB threshold. Now, since the IF bandwidth required is twice the modulation bandwidth

$$N_{IF} = 2kTb$$

and since

$$C = \frac{P_T}{L}$$

then $\frac{P_T}{L} \geq 32kTb$ or 15.1 dB above a power kTb .

It follows from the above that if

$$\left(\frac{S}{N}\right) = 10^4 (40\text{dB})$$

then $M = 14.5$, and a 5-mc signal bandwidth requires a 155-mc RF bandwidth. To accomplish this exactly requires infinite feedback, but for practical purposes a feedback factor of 145 (43 dB), reducing the modulation index in the IF to 0.1, should suffice.

The transmitter power can be kept constant and the output signal-to-noise ratio increased (up to the limit given by information theory) by further increasing in equal amounts the transmitter index, receiver feedback, and bandwidth in the medium.

Consider what this implies in terms of an actual system.

A. FM Feedback — Passive Repeater

Assuming a maser amplifier is in use, the receiver noise becomes negligible. The sky temperature is then the dominant factor and, referring to Fig. 3, choose 2000 mc for the operating frequency; assuming that the antennas point at elevations no lower than 7.25 °K, so that the noise is 13 dB above that at 1 °K. With no more than 1% interruption of service, this requires 24 satellites at orbits 3000 mi high. The maximum distance (rms) is 5420 mi (Table 3). The shortest distance, that is, when the satellite is half-way between the terminals, is 3300 mi (Table 2).

Suppose the antenna diameters are 150 ft, and efficiencies 60%. Further, assume that the passive repeater is a metallic sphere of 100-ft diameter. Thus, the maximum path loss:

$$L = \frac{16 \times (1/2)^2 \times (5240)^4 \times (5280)^4}{(\pi \times 75^2)^2 \times (0.60)^2 \times (100)^2}$$

$$L = 2.04 \times 10^{18} \text{ or } 183.1\text{dB}$$

The minimum loss turns out to be 8 dB less.

For 1-Hz baseband bandwidth and 1 °K, the received power must be 15.1 dB above the thermal level of -228.9 dBw. A temperature of 20 °K increases the required power

by 13.0 dB, and a bandwidth b of 5 mc increases the power by 67.0 dB.

In dB-watt

$$P_T = +183.1 + 15.1 - 228.9 + 13 + 67 \text{ dBw}$$

$$P_T = 49.3 \text{ dBw or } 85 \text{ kw.}$$

This amount of CW power can, in principle, be supplied with existing tubes (Varian Associates VA-800, 1.7 to 2.4 kmc, 10-kw klystrons), either of them drive in parallel.

Suppose a higher frequency is chosen, for example, 8000 mc. This reduces the power requirement by a factor

$$\left(\frac{8000}{2000}\right)^2 = 16$$

On the other hand, the antenna noise temperature will be increased by a few degrees, e.g., from 20 to 25°K, giving a factor 1.25. We assume that antenna size and efficiency are unchanged, which actually implies a considerable increase in the cost of the antennas, and arrive at a transmitter power of

$$P_T = \frac{85 \times 1.25}{16} = 6.65\text{kw}$$

This amount of power can be supplied by four Klystrons in parallel, such as the Varian Associates VA-806. Further, it appears that single tubes could be developed for either of the cases discussed above.

B. FM Feedback — Active Repeater

In the active repeater it is assumed that there is a wide-band low-noise RF amplifier with a power output limited to P_α by weight, life, and bulk considerations. It is connected to the earth terminals by means of nondirectional antennas, namely dipoles with effective areas $3\lambda^2/8\pi$.

It can be shown that the crucial factors are the size and noise temperature of the antenna at the receiving terminal and the satellite output power. The transmitter power on the ground and the noise figure of the satellite amplifier are of minor importance.

The path loss of importance now is given by

$$\frac{P_R}{P_\alpha} = \frac{A_{sat} A_R \cdot \eta}{\lambda^2 p^2} = \frac{1}{L}$$

where

$$A_{sat} = 3\lambda^2/8\pi$$

$$A_R = \pi\alpha^2/4, \alpha \text{ being the antenna diameter}$$

$$\eta = \text{antenna efficiency}$$

$$p = \text{path length.}$$

Hence

$$L = \frac{32}{3} \left(\frac{p^2}{\alpha^2 \eta} \right)$$

Employing FM with feedback as before, use the formula

$$\frac{P_\alpha}{L} = 32kTb$$

Thus, the necessary RF power at the satellite

$$P_{\alpha} = 326 \frac{p^2}{\alpha^2 \eta} kTb$$

Take the same antenna, frequency, noise temperature, bandwidth, and distance as used before in the passive repeater case, namely

$$\begin{aligned} p &= 5240 \text{ mi} \\ \alpha &= 150 \text{ ft} \\ \eta &0.6 \\ T &= 20^{\circ}\text{K at } 2000 \text{ mc} \\ b &= 5 \text{ mc.} \end{aligned}$$

This requires a satellite transmitter power of $P_{\alpha} = 25.4$ mw.

This does not seem to be much in the way of RF power; an increase of one order of magnitude is perhaps not out of the question. Overall efficiencies of something like a few percent may perhaps be achieved, leading to a continuous power requirement of a few watts. This can be satisfied by a combination of solar and storage batteries, but owing to the limited life of existing storage batteries, the life of such a repeater would at present be restricted.

An active repeater in a 24-hour orbit with the ground installations much as described above will have to put out approximately $(25\,000/500)^2 = 25$ times the transmitter power calculated above, that is, about 625 mw. This is more than enough, since a 24-hour satellite will, in all probability, be attitude stabilized and therefore could carry a high-gain antenna to great advantage.

The power available on this kind of repeater has to include provision for the maintenance of accurate position, velocity, and attitude; how much, it is difficult to estimate at this time. Life will again be a serious problem.

VI. MODULATION SYSTEMS

Microwave tubes such as klystrons, traveling-wave tubes, and backward-wave oscillators lend themselves very conveniently to frequency modulation. Magnetrons and amplitrons are operated advantageously under pulse conditions.

Ruthroff and Goodall have shown [3] that FM with feedback and PPM give practically the same performance when the same mean transmitter power is employed. It would not be surprising if other modulation schemes were to be found to give similar performance when all requisite conditions are optimized.

All modulation systems will have to operate in the presence of large and continuously varying Doppler-frequency shifts. With approximately spherical reflectors at an elevation higher than a few degrees above the horizon, no fading, scintillating, or glinting is expected to occur; thus, no fading margin was deemed to be necessary in the path-loss calculations. The noise temperature, antenna size, power, etc., are the most reasonable estimate that can be made at present of what is necessary to provide the

performance specified. In an actual system, a greater power might be used in order to provide a margin of safety.

VII. UNKNOWNNS IN SATELLITE COMMUNICATIONS

The authors have a great deal of confidence in the overall feasibility of satellite communications. Nevertheless, quite a few unknowns exist. Experiments, development, and experience are needed before all problems can be considered solved. Some problems are of a fundamental nature, such as the influence of the earth's atmosphere on the propagation of radiation; others are instrumental, such as the limits on receiver noise temperature due to losses and mismatches.

A. Satellite Construction

The passive repeater envisioned in Section V-A is a metallized plastic sphere of 100-ft diameter. A considerable amount of work is being done by the National Aeronautics and Space Agency, who have announced that they will place one or more balloons of this type into orbit sometime in 1959. The major unknown at present is the life of such balloons. Also in question is the ultimate shape.

Other satellite constructions, such as metallic wire-mesh spheres, the mesh being small compared with a wavelength, have been suggested, and methods of placing them in orbits deserve to be studied.

A spherical satellite scatters the radiation which it incepts isotropically. Satellites of shapes other than spherical could be used to reflect a greater fraction of the radiation striking them from the transmitter to the receiver.

A great deal of work needs to be done on active repeaters, particularly on components such as microwave tubes, storage batteries, capacitors, etc., before a sufficiently long life can be assured. Many components may have to be constructed on an entirely new basis, taking into account that the environment will be radically different from any encountered so far, namely, ultrahigh vacuum, intense ultraviolet, X-ray and cosmic ray bombardment, and micrometeorites.

B. Propagation Effects

The whole of the earth's atmosphere will have to be traversed twice in every satellite radio link; it is therefore important to know the effect it will have on the beams of microwave radiation. It is fairly certain that nothing harmful will happen to beams pointing at more than 10° above the horizon, except perhaps when traversing auroral regions. nevertheless, propagation measurements are required.

It would be desirable to know the actual instantaneous angle of arrival of beams of radiation coming from all possible directions over a wide frequency range and covering a wide range of climatic conditions. Data concerning rotation of the plane of polarization are also important. The effective sky temperature as a function of frequency and elevation must be ascertained.

C. Antenna Considerations

Some of the research results of the preceding section will determine the largest size of antenna which can be used with advantage.

Another problem to be solved is that of an antenna with low effective noise temperature; that is, one in which the losses and the side and back lobes have been reduced below a certain tolerable limit.

Problems of very large steerable antennas call for solution; these may be divided into electrical and mechanical problems. It may be that both will be solved eventually by adopting the principles of multielement steerable arrays.

D. Overall System Noise Figure

This is largely an RF input problem, once the antenna itself has been "cleaned up." With very large antennas it will be necessary, for economical and mechanical reasons, to combine into one unit both transmitter and receiver feed horns and also perhaps the output and input of tracking radar. The frequencies employed for all these functions may be spaced widely apart so that efficient filtering should not be too difficult.

Nevertheless, to achieve an effective low-noise temperature will require much competent and painstaking experimentation. Considerable development work is already in progress on masers and parametric amplifiers. Not so much has been done on tying them in with a particular communication system.

E. Tracking of Satellites

Satellites move in smooth, regular orbits, predictable with high precision. This makes it attractive to think of using computers, analog or digital, for the purpose of steering antennas on them.

The alternative method employs a tracking radar. For relatively small antennas, or in case only a feed system has to move, the tracking radar may have a separate antenna, and the communication antenna is "slaved" to the radar.

With large antennas, which may distort, sag, or twist as they are slewed about or in the presence of high winds, it might be necessary to make the radar output and input integral with the communication feed system in order to point the antenna accurately despite distortions with respect to the mounting and drive.

Similar considerations also apply to the Doppler-shift of the reflected radiation, which can be computed beforehand, or which can be derived instantaneously from the radar data.

The results of the research on propagation effects will affect solutions to the tracking problems. Any satellite communication system involving very large antennas at microwave frequencies will depend entirely on an accurate and dependable tracking system such as probably has never been built before.

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